**Propositional rules: Axioms, basic actions, conditionals, switching, composition.**

$$\bot \langle \boldsymbol{a} \mid - \rangle P \quad P_\exists \langle - \mid \boldsymbol{\alpha} \rangle \top \quad P_\exists \langle \lambda \boldsymbol{x}.\boldsymbol{x} \mid - \rangle P_\exists \quad \frac{P_\exists \to Q_\forall \in \mathrm{Ax}}{P_\exists \langle - \mid - \rangle Q_\forall} \quad \frac{P_\exists \langle - \mid - \rangle Q_\forall}{P'_\exists \langle - \mid - \rangle Q'_\forall} \text{ for } \frac{P_\exists \to Q_\forall}{P'_\exists \to Q'_\forall} \in \mathrm{Rule}$$

$$\frac{P_\exists \langle \boldsymbol{a}, \boldsymbol{b} \mid - \rangle Q \wedge R}{P_\exists \langle \boldsymbol{b}, \boldsymbol{a} \mid - \rangle R \wedge Q}\, {}^{p\wedge R} \qquad \frac{P_\exists \wedge Q_\exists \langle \boldsymbol{a} \mid - \rangle R}{Q_\exists \wedge P_\exists \langle \tilde{\boldsymbol{a}} \mid - \rangle R}\, {}^{p\wedge L} \qquad \frac{P_\exists \langle \boldsymbol{a}, \boldsymbol{b} \mid - \rangle Q \vee_c R}{P_\exists \langle \boldsymbol{b}, \boldsymbol{a} \mid - \rangle R \vee_{\tilde{c}} Q}\, {}^{p\vee R} \qquad \frac{P_\exists \vee_c Q_\exists \langle \boldsymbol{a} \mid - \rangle R}{Q_\exists \vee_{\tilde{c}} P_\exists \langle \tilde{\boldsymbol{a}} \mid - \rangle R}\, {}^{p\vee L}$$

$$\frac{P_\exists \langle \boldsymbol{a} \mid - \rangle Q}{P_\exists \langle \boldsymbol{a}, \boldsymbol{b} \mid - \rangle Q \vee_0 R}\, {}^{\vee R} \qquad \frac{P_\exists \langle \boldsymbol{a} \mid - \rangle Q}{P_\exists \wedge R_\exists \langle \boldsymbol{a}_\pi \mid - \rangle Q}\, {}^{\wedge L} \qquad \frac{P_\exists \langle \boldsymbol{a}, \boldsymbol{b} \mid - \rangle Q \wedge R}{P_\exists \langle \boldsymbol{a} \mid - \rangle Q}\, {}^{\wedge R} \qquad \frac{P_\exists \vee_0 R_\exists \langle \boldsymbol{a} \mid - \rangle Q}{P_\exists \langle \boldsymbol{a}_p \mid - \rangle Q}\, {}^{\vee L}$$

$$\frac{P_\exists \wedge \phi \langle \boldsymbol{a} \mid - \rangle R \quad Q_\exists \wedge \neg\phi \langle \boldsymbol{b} \mid - \rangle R}{P_\exists \vee_\phi Q_\exists \langle \lambda \boldsymbol{x}, \boldsymbol{y}.\mathtt{if}(\phi, \boldsymbol{a}\boldsymbol{x}, \boldsymbol{b}\boldsymbol{y}) \mid - \rangle R}\, {}^{cond_L} \qquad \frac{P_\exists \langle \boldsymbol{a} \mid - \rangle Q \quad P_\exists \langle \boldsymbol{b} \mid - \rangle R}{P_\exists \langle \boldsymbol{a}, \boldsymbol{b} \mid - \rangle Q \wedge R}\, {}^{cond_R}$$

$$\frac{P_\exists \langle \boldsymbol{a} \mid - \rangle Q_\exists \to R}{P_\exists \wedge Q_\exists \langle \boldsymbol{a} \mid - \rangle R}\, {}^{imp} \qquad \frac{P_\exists \wedge Q_\exists \langle \boldsymbol{a} \mid - \rangle R}{P_\exists \langle \boldsymbol{a} \mid - \rangle Q_\exists \to R}\, {}^{exp} \qquad \frac{P_\exists \langle \boldsymbol{a} \mid - \rangle Q_\exists \quad Q_\exists \langle \boldsymbol{b} \mid - \rangle R}{P_\exists \langle \boldsymbol{b} \circ \boldsymbol{a} \mid - \rangle R}\, {}^{comp}$$

**Quantifier rules: Term introduction, $\boldsymbol{\lambda}$-abstraction and application, epsilon terms.**

$$\frac{P_\exists \langle \boldsymbol{a} \mid - \rangle Q(\boldsymbol{t})}{P_\exists \langle \lambda_.t, \boldsymbol{a} \mid - \rangle \exists \boldsymbol{x}\, Q(\boldsymbol{x})}\, {}^{\exists_R}$$

$$\frac{P_\exists(\boldsymbol{x}) \langle \boldsymbol{a} \mid - \rangle Q}{\exists \boldsymbol{x}\, P_\exists(\boldsymbol{x}) \langle \lambda \boldsymbol{x}.\boldsymbol{a} \mid - \rangle Q}\, {}^{\exists_L} \qquad \frac{P_\exists \langle \boldsymbol{a} \mid - \rangle Q(\boldsymbol{x})}{P_\exists \langle \lambda \boldsymbol{y}, \boldsymbol{x}.\boldsymbol{a}\boldsymbol{y} \mid - \rangle \forall \boldsymbol{x}\, Q(\boldsymbol{x})}\, {}^{\forall_R} \qquad \boldsymbol{x} \text{ not free in } Q \text{ resp. } P \text{ for } \exists_L \text{ resp. } \forall_R$$

$$\frac{\exists \boldsymbol{x}\, P_\exists(\boldsymbol{x}) \langle \boldsymbol{a} \mid - \rangle Q}{P_\exists(\boldsymbol{t}) \langle \boldsymbol{a}\boldsymbol{t} \mid - \rangle Q}\, {}^{s_L} \qquad \frac{P_\exists \langle \boldsymbol{a} \mid - \rangle \forall \boldsymbol{x}\, Q(\boldsymbol{x})}{P_\exists \langle \lambda \boldsymbol{y}.\boldsymbol{a}\boldsymbol{y}\boldsymbol{t} \mid - \rangle Q(\boldsymbol{t})}\, {}^{s_R} \qquad \frac{P_{qf} \langle \boldsymbol{a}, \boldsymbol{b} \mid - \rangle \exists \boldsymbol{x}\, Q(\boldsymbol{x})}{P_{qf} \langle \boldsymbol{b} \mid - \rangle Q(\boldsymbol{a})}\, {}^{\epsilon_R}$$

**Consequence, extensionality, and induction/recursion**

$$\frac{P'_\exists \to_D P_\exists \quad P_\exists \langle \boldsymbol{a} \mid - \rangle Q \quad Q \to_D Q'}{P'_\exists \langle \boldsymbol{a} \mid - \rangle Q'}\, {}^{cons} \qquad \frac{P_\exists \langle \boldsymbol{a} \mid \boldsymbol{\alpha} \rangle Q \quad \boldsymbol{a}, \boldsymbol{\alpha} = \boldsymbol{b}, \boldsymbol{\beta}}{P_\exists \langle \boldsymbol{b} \mid \boldsymbol{\beta} \rangle Q}\, {}^{ext} \qquad \frac{P_\exists(x) \langle \boldsymbol{a}x \mid - \rangle P_\exists(x+1)}{P_\exists(0) \langle \mathtt{rec}\, \boldsymbol{a} \mid - \rangle \forall x\, P_\exists(x)}\, {}^{ind}$$